GENERAL

1. THE ELECTRICAL INSTALLATION SHALL COMPLY WITH LILLY SPECIFICATIONS, CONSTRUCTION DRAWINGS, THE 2009 EDITION OF THE INDIANA STATE AND LOCAL ELECTRICAL CODES, INDIANA OSHA (IOSHA) REQUIREMENTS, AND THE 2023 NATIONAL ELECTRICAL CODE (NEC). IF CONFLICTING INFORMATION IS FOUND BETWEEN PROJECT DRAWINGS, AND/OR APPLICABLE STANDARDS AND CODES, THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONSTRUCTION MANAGER FOR RESOLUTION AND THE MORE STRINGENT OF THE REQUIREMENTS SHALL APPLY.
2. ELECTRICAL MATERIALS SHALL BE SUITABLE FOR THE APPLICATION AND APPROVED FOR THE USE. EQUIPMENT REQUIRED TO BE LISTED, LABELED, OR BOTH SHALL LISTED OR LABELED BY A NATIONAL RECOGNIZED TESTING LABORATORY (NRTL) AND INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS INCLUDED IN THE LISTING, OR LABELING.
3. IF THERE IS A DISCREPANCY BETWEEN THE GENERAL NOTES AND THE PROJECT SPECIFICATION, CONSULT WITH THE CONSTRUCTION MANAGER.
4. THE ELECTRICAL INSTALLATION SHALL BE DONE IN A NEAT, SKILLFUL, AND PROFESSIONAL MANNER AS DESCRIBED IN ANSI/NECA 1, STANDARD FOR GOOD WORKMANSHIP IN ELECTRICAL CONSTRUCTION.
5. PLANS AND DETAILS ARE DIAGRAMMATIC, UNLESS NOTED OTHERWISE. THE ELECTRICAL CONTRACTOR SHALL INSTALL EQUIPMENT AS CLOSE AS PRACTICABLE TO THE LOCATIONS INDICATED ON THE DRAWINGS. GENERALLY, MINOR DEVIATIONS DUE TO FIELD CONDITIONS SHALL BE PERMITTED. MAJOR DEVIATIONS FROM THE DESIGN SHALL NOT BE ALLOWED WITHOUT PRIOR ENGINEERING APPROVAL.
6. WHERE FINAL EQUIPMENT INSTALLATION LOCATIONS VARY FROM THE CONSTRUCTION DRAWINGS, THE FINAL LOCATIONS SHALL BE DOCUMENTED (AS-BUILT) ON A SINGLE MARKED SET OF DRAWINGS AND RETURNED TO THE CONSTRUCTION MANAGER AFTER CONSTRUCTION IS COMPLETE.
7. THE ELECTRICAL CONTRACTOR SHALL COORDINATE THE INSTALLATION OF ELECTRICAL EQUIPMENT WITH OTHER TRADES IN ORDER TO AVOID POTENTIAL INTERFERENCES. IF INTERFERENCES OCCUR, THE ELECTRICAL CONTRACTOR SHALL NOTIFY THE CONSTRUCTION MANAGER IMMEDIATELY FOR RESOLUTION.
8. FIELD WELDING ON STRESS RELIEVED (HEAT TREATED AFTER FABRICATION) TOWERS, STRUCTURAL STEEL, VESSELS OR TANKS IS PROHIBITED.
9. WHERE THE CORROSION PROTECTIVE FINISH ON ANY MATERIAL OR EQUIPMENT HAS BEEN REMOVED BY MACHINING, WELDING, DRILLING, ETC. OR ACCIDENTALLY DAMAGED, THE ORIGINAL FINISH SHALL BE RESTORED BY APPLYING AN APPROVED EPOXY OR ZINC PAINT.
10. A MINIMUM HEIGHT OF 6'-6" OR THE HEIGHT OF EQUIPMENT, WHICHEVER IS GREATER SHALL BE MAINTAINED IN THE WORKING SPACE ABOUT ELECTRICAL EQUIPMENT AND ACCESS/EGRESS AISLES, UNLESS NOTED OTHERWISE.
11. ALL UNUSED CONDUIT ENTRIES IN FITTINGS, BOXES, AND EQUIPMENT SHALL BE CLOSED OR PLUGGED IN AN APPROVED MANNER TO MAINTAIN THE LISTING AND/OR NEMA ENCLOSURE RATING.
12. ELECTRICAL EQUIPMENT SUPPORTS SHALL BE INSTALLED IN A MANNER THAT WILL NOT WEAKEN THE STRUCTURE IT IS BEING ATTACHED TO. THE CONTRACTOR SHALL NOT MODIFY, CUT, BURN, OR DRILL ANY STRUCTURAL MEMBER NOT INDICATED ON THE DRAWINGS WITHOUT WRITTEN APPROVAL FROM ENGINEERING.
13. GENERAL FASTENING HARDWARE, (I.E., BOLTS, SCREWS, NUTS, WASHERS, CONDUIT STRAPS, ETC.), USED IN OUTDOOR, HYGIENIC, OR CORROSIVE ENVIRONMENTS SHALL BE 316 STAINLESS STEEL, UNLESS NOTED OTHERWISE. ELECTROGALVANIZED FINISHED HARDWARE IS PERMITTED FOR INDOOR, NON-CORROSIVE LOCATIONS SUCH AS OFFICES, CONFERENCE ROOMS, ABOVE CEILINGS, ETC., GROUNDING HARDWARE SHALL BE SILICON BRONZE, UNLESS NOTED OTHERWISE. NOTE 13 REVISION IS NOT RETROACTIVE FOR HARDWARE ALREADY INSTALLED BUT SUPERSEDES MATERIAL TYPES INSTALLED AFTER NOVEMBER 20, 2025, REVISION 5.
14. MINIMUM WORKING SPACE ABOUT ELECTRICAL EQUIPMENT 1000 V NOMINAL, OR LESS, SHALL BE IN ACCORDANCE WITH 2023 NEC, ARTICLE 110.
15. MINIMUM WORKING SPACE IN MANHOLES AND OTHER ELECTRICAL ENCLOSURES INTENDED FOR PERSONNEL ENTRY SHALL COMPLY WITH NEC ART. 110, PART V.
16. THE ELECTRICAL CONTRACTOR SHALL PURCHASE AND INSTALL ELECTRICAL MATERIALS AND EQUIPMENT IN ACCORDANCE WITH PROJECT SPECIFICATIONS AND DRAWINGS.ALL ELECTRICAL EQUIPMENT PROVIDED SHALL BE NEW. RECONDITIONED EQUIPMENT SHALL BE PROHIBITED.
17. MISCELLANEOUS ELECTRICAL EQUIPMENT SUPPORTS SHALL BE DESIGNED AND FURNISHED BY THE ELECTRICAL CONTRACTOR, UNLESS NOTED OTHERWISE.
18. ELECTRICAL EQUIPMENT SUBJECT TO MOISTURE EITHER INDOORS OR OUTDOORS SHALL BE LOCATED ON CONCRETE PADS 3 1/2" THICK (MIN.) PROVIDED BY CIVIL, UNLESS NOTED OTHERWISE.
19. ELECTRICAL EQUIPMENT, LUMINAIRES, DEVICES SUCH AS SWITCHES AND RECEPTACLES, CABLE TRAY, CABLES, BOXES, PULL BOXES, ETC., SHALL HAVE NAMEPLATES SUPPLIED AND INSTALLED BY THE ELECTRICAL CONTRACTOR IN ACCORDANCE WITH LILLY SPECIFICATION LP100-265-EMS-16195.
20. THE ELECTRICAL CONTRACTOR SHALL OBTAIN APPROVALS FROM THE CONSTRUCTION MANAGER BEFORE PUNCHING HOLES OR CUTTING FINISHED SURFACES, OR BUILDING STEEL. SUBSEQUENT REPAIRS SHALL BE PERFORMED BY APPROPRIATELY SKILLED CRAFTSMEN. ALL REPAIRS MADE TO FINISHED SURFACES SHALL MATCH THE EXISTING FINISH.
21. BRANCH CIRCUITS FROM DIFFERENT SOURCES SHALL NOT BE ROUTED IN THE SAME CONDUIT, CABLE TRAY, CHANNEL, OR RACEWAY, (I.E., EMERGENCY CIRCUITS AND NORMAL CIRCUITS, DIFFERENT PANEL BOARDS, ETC.), UNLESS NOTED OTHERWISE.
22. CONDUITS SHALL ENTER ENCLOSURES FROM THE BOTTOM IN WET OR OUTDOOR LOCATIONS. IF NECESSARY DUE TO LACK OF SPACE ON THE BOTTOM, CONDUITS MAY ENTER NON-EXPLOSION ENCLOSURES ON THE SIDE IF LOW POINT CONDUIT SYSTEM DRAINS ARE INSTALLED AND APPROVED WATERTIGHT CONDUIT HUBS SUCH AS MYERS OR EQUAL ARE USED.
23. THE ELECTRICAL CONTRACTOR SHALL KEEP WORK AREAS FREE OF DEBRIS.
24. ELECTRICAL EQUIPMENT ROOMS SHALL BE KEPT REASONABLY CLEAN FROM DUST AND DIRT. FOOD AND DRINKS ARE PROHIBITED IN ELECTRICAL ROOMS AND RODENT MITIGATION/PREVENTION IS REQUIRED DURING CONSTRUCTION.
25. THE ELECTRICAL CONTRACTOR SHALL NOT UNDERTAKE ANY WORK THAT WILL INTERFERE WITH, DELAY, OR HALT OPERATIONS OF THE SITE WITHOUT WRITTEN APPROVAL FROM THE CONSTRUCTION MANAGER.
26. POWER AND LIGHTING CIRCUITS SHALL BE SEGREGATED AND SPACED IN ACCORDANCE WITH THE MINIMUM SPACING DISTANCES PROVIDED ON DWG. LP100-EJ-00000007. IF THERE ARE DISCREPANCIES BETWEEN THE DRAWINGS AND THE 3D MODEL, THE CONTRACTOR SHALL CONSULT WITH ENGINEERING BEFORE PROCEEDING.

ABOVEGROUND CONDUIT

1. THE TERM "CONDUIT" WITHIN THESE NOTES INCLUDE RACEWAYS SUCH AS RIGID METAL, FIBERGLASS, PVC, ELECTRICAL METALLIC TUBING (EMT), FLEXIBLE CONDUIT, ETC., UNLESS NOTED OTHERWISE.
2. CONDUIT SYSTEMS SHALL BE SECURELY FASTENED WITH STRAPS IDENTIFIED FOR THE PURPOSE OR OTHER APPROVED MEANS.
3. ABOVEGROUND CONDUIT SHALL BE 3/4" (MINIMUM), UNLESS NOTED OTHERWISE OR SPECIFIED ON THE DRAWINGS. CONDUIT BODIES FOR RMC OR STAINLESS STEEL SHALL BE FORM 8, UNLESS NOTED OTHERWISE.
4. CONDUIT BODIES INSTALLED IN CLEAN ROOMS SHALL BE 316 SS, UNLESS NOTED OTHERWISE.
5. EMT WITH DIE-CAST COMPRESSION TYPE FITTINGS MAY BE USED IN DRY, INDOOR, NON-HAZARDOUS, NON-CORROSIVE ENVIRONMENTS, AND WHERE NOT SUBJECT TO PHYSICAL DAMAGE OR VIBRATION. EMT SHALL BE USED IN OFFICE AREAS, CONFERENCE ROOMS, BATHROOMS, I.E., LIGHT COMMERCIAL INSTALLATIONS BUT NOT IN PRODUCTION OR INDUSTRIAL PORTIONS OF THE FACILITY UNLESS NOTED OTHERWISE.
6. CONDUIT AND FITTINGS INSTALLED IN CLEAN ROOMS SHALL BE STAINLESS STEEL, UNLESS NOTED OTHERWISE. EXPLOSIONPROOF CONDUIT SEALS LARGER THAN 1" SHALL BE MALLEABLE IRON ALLOY AND PAINTED WITH APPROVED FDA APPROVED PAINT AFTER FIBER DAM IS PLACED AND SEAL IS POURED. DRAINS ON SEALS SHALL NOT BE PAINTED.
7. LIQUIDTIGHT FLEXIBLE METAL CONDUIT (LFMC) SHALL BE SUITABLE FOR THE LOCATION WHERE REQUIRED TO CONNECT EQUIPMENT REQUIRING OCCASIONAL REMOVAL FROM SERVICE, OR SUBJECT TO MINOR VIBRATION. WHERE USED IN CLEAN ROOMS LFMC SHALL BE FDA APPROVED. FLEXIBLE CONDUIT LENGTHS SHALL BE AS SHORT AS PRACTICAL AND SUPPORTED PER THE NEC.
8. EXPLOSIONPROOF CONDUIT SEALS SHALL BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. SEALS THAT HAVE BEEN POURED WITH SEALANT SHALL BE PERMANENTLY IDENTIFIED IN AN APPROVED MANNER.
9. EXPLOSIONPROOF CONDUIT SEALS, WITH OR WITHOUT DRAINS, SHALL BE THE EXPANDED FILL TYPE ALLOWING FOR MAXIMUM 40% CONDUIT FILL OF THE RACEWAY.
10. ALL CONDUIT ENDS SHALL BE CUT SQUARE, REAMED SMOOTH WITH NO BURRS, AND DO NOT CONTAIN FOREIGN MATERIALS. WHERE THREADED, SHALL BE MADE WRENCHTIGHT. ALL RACEWAYS SYSTEMS SHALL BE COMPLETE BEFORE THE INSTALLATION OF WIRES OR CABLES. EMPTY RACEWAY ENDS SHALL BE PLUGGED UNTIL WIRE OR CABLE IS INSTALLED.
11. MINIMUM CABLE BENDING RADIUS SHALL NOT EXCEED PERMITTED BY THE CABLE MANUFACTURER AND THE NATIONAL ELECTRICAL CODE. OVERSIZED CONDUIT BODIES AND MOGUL BODIES SHALL BE USED ON ALL CONDUIT RUNS, WHERE NECESSARY, SO THE CABLE BENDS DO NOT EXCEED THE MINIMUM BENDING RADII.
12. CONDUITS INSTALLED OUTDOORS OR IN WET LOCATIONS SHALL HAVE LOW POINT DRAINS INSTALLED IN ALL CONDUIT RUNS BEFORE ENTERING ELECTRICAL ENCLOSURES OR WHERE SPECIFIED ON INSTALLATION DETAILS.
13. CONDUIT EXPANSION FITTINGS SHALL BE INSTALLED IN STRAIGHT RUNS OF CONDUIT LOCATED OUTDOORS IN ACCORDANCE WITH NEC 300.7(B). A METAL TEMPERATURE DIFFERENTIAL OF 145 DEG. F SHALL BE USED AS THE BASIS FOR CALCULATIONS IN DETERMINING THE QUANTITY OF EXPANSION FITTINGS REQUIRED.
14. FACTORY CONDUIT ELBOWS MAY BE USED FOR CONDUIT TRADE SIZES 2" AND LARGER AND UNDERGROUND.
15. CONDUCTIVE, ANTI-SEIZE LUBRICANT IDENTIFIED FOR THE PURPOSE SHALL BE APPLIED TO THREADED CONDUIT TERMINATIONS AND THREADED CONDUIT BODY COVER SCREWS.
16. THE ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL APPROVED FIRE RATED PENETRATIONS FOR CONDUITS PASSING THROUGH FLOORS, WALLS, AND ROOFS. SLEEVES SHALL BE FABRICATED FROM STEEL PIPE, WHICH SHALL BE AT LEAST ONE NOMINAL PIPE SIZE LARGER THAN THE CONDUIT. GENERALLY, FLOOR SLEEVES SHALL PROTRUDE 4" ABOVE THE FINISHED FLOOR, AND ROOF SLEEVES SHALL PROTRUDE 12" ABOVE THE ROOF DECK. PATCHING AROUND SLEEVES SHALL BE PERFORMED BY APPROPRIATELY SKILLED CRAFTSMEN, AND PATCHING MATERIALS SHALL MATCH THE BUILDING FINISH. SEE DWG. LP100-EJ-00000413.
17. PROCESS PIPING OR OTHER ELECTRICAL RACEWAYS SHALL NOT SUPPORT CONDUIT, EMT, OR EQUIPMENT UNLESS INDICATED ON INSTALLATION DETAILS AND COMPLY WITH THE NEC.
18. ROUTE ABOVEGROUND CONDUITS PARALLEL OR PERPENDICULAR TO STRUCTURAL COLUMNS OR STRUCTURAL STEEL FRAMING MEMBERS UNLESS INDICATED OTHERWISE ON PLAN DRAWINGS. CANTILEVERED FIELD SUPPORTS SHALL NOT EXCEED 18 IN. FROM THE CENTER OF THE FARTHEST CONDUIT TO THE CENTER OF THE STRUCTURAL STEEL WITHOUT PRIOR ENGINEERING APPROVAL.
19. CONDUIT BODIES SHALL BE INSTALLED IN A MANNER THAT THE COVERS CAN BE EASILY REMOVED AND THE WIRING IS ACCESSIBLE.
20. CONDUIT ROUTED BEHIND LADDERS SHALL BE INSTALLED IN SUCH A MANNER THAT THEY WILL NOT CAUSE A FALSE OR INSECURE STEP. A MINIMUM OF 7 IN. OF CLEAR SPACE SHALL BE MAINTAINED BETWEEN THE BACK SIDE OF THE LADDER RUNGS AND ANY OBSTRUCTIONS.
21. METALLIC RACEWAYS SHALL BE BONDED IN ACCORDANCE WITH THE NEC. WHERE RMC TRANSITIONS TO METALLIC CABLE TRAY, LISTED BONDING CONDUIT CONNECTORS, SUCH AS CROUSE HINDS TYPE LCC CLAMPS ARE PERMITTED IN LIEU OF COPPER WIRE BONDING JUMPERS, GROUNDING BUSHINGS AND LUGS.

ABOVEGROUND CONDUIT CONT'D

22. METAL CLAD CABLE (TYPE MC) SHALL NOT BE INSTALLED IN CONDUIT, UNLESS PRIOR ENGINEERING APPROVAL IS OBTAINED AND THE METAL JACKET SHALL NOT BE USED AS THE SOLE EQUIPMENT GROUNDING CONDUCTOR.
23. CONDUITS SHALL BE TAGGED AS INDICATED ON DWG. LP100-EJ-00000010.

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24. PER SPECIFICATION LP100-265-EMS-16100, BASIC ELECTRICAL MATERIAL AND METHODS, PART 3, SECTION 3.01.A.2.B., LOCATIONS SUBJECT TO PHYSICAL DAMAGE SHALL BE RIGID METAL CONDUIT (TYPE RMC) WHERE RACEWAY WIRING METHODS ARE USED. ALL EXPOSED CONDUIT, CABLE, OR TUBING SHALL REQUIRE RMC EXTENDING FROM AN OUTLET OR EQUIPMENT UP TO NOT LESS THAN 10'-0" (MIN.) MEASURED FROM GRADE OR FINISHED FLOOR, PLATFORM, INTERSTITIAL SPACE CEILING/FLOOR, ETC., FOR PHYSICAL PROTECTION. AT ELEVATIONS MORE THAN 10'-0" ABOVE THESE SURFACES, OTHER TYPES OF RACEWAYS, INCLUDING EMT, MAY BE USED IN ACCORDANCE WITH THE NEC AND THE REFERENCED SPECIFICATION SECTION 3.01.A.2.D. CONCEALED RACEWAYS AND CABLES, (I.E., CONDUIT, TUBING, METAL CLAD CABLE, ETC.) INSIDE STRUCTURAL WALLS ARE NOT SUBJECT TO THIS REQUIREMENT.

GROUNDING

1. FOR STANDARD GROUNDING INSTALLATION DETAILS, SEE DRAWING # LP100-EJ-00000100 SERIES DRAWINGS.
2. GROUNDING TAPS SHALL BE INDIVIDUALLY CONNECTED TO THE MAIN GROUNDING RING CONDUCTOR. TAPS FROM TAPS ARE PROHIBITED.
3. UNDERGROUND GROUNDING CONDUCTOR TAPS SHALL BE OF SUFFICIENT LENGTH TO EXTEND 10'-0" (MIN.) ABOVE FINISHED GRADE UNLESS NOTED OTHERWISE AND SHALL BE PHYSICALLY PROTECTED DURING CONSTRUCTION.
4. ALL BELOW GRADE GROUNDING SYSTEM CONDUCTORS SHALL BE BURIED A MINIMUM OF 30 INCHES IN TRENCHES WITH SLACK BETWEEN CONNECTIONS, UNLESS NOTED OTHERWISE.
5. RESISTANCE TO EARTH OF THE COMPLETED GROUND SYSTEM SHALL NOT EXCEED 1 OHM (MAX.), UNLESS NOTED OTHERWISE. THE ELECTRICAL CONTRACTOR SHALL VERIFY THE RESISTANCE VALUE BY TESTING AFTER THE GROUNDING ELECTRODE SYSTEM IS COMPLETELY INSTALLED. IF MORE THAN 1 OHM IS MEASURED, ADDITIONAL GROUND RODS OR GROUNDING ENHANCEMENT MATERIALS SHALL BE INSTALLED TO ACHIEVE A MAXIMUM RESISTANCE TO EARTH OF 1 OHM.
6. EXOTHERMIC WELDING SHALL NOT BE USED UNLESS INDICATED ON THE DRAWINGS.
7. THE GROUNDING SYSTEM SHALL BE CONNECTED TO UNDERGROUND PIPING ONLY WHERE INDICATED TO AVOID INTERFERENCE WITH CATHODIC PROTECTION SYSTEMS.
8. ABOVEGROUND GROUNDING CONNECTIONS OTHER THAN CORROSIVE LOCATIONS OR CLEANROOMS SHALL BE MADE WITH LISTED COMPRESSION LUGS OR BOLTED MECHANICAL CONNECTORS WITH SILICON BRONZE HARDWARE TO ENSURE A PERMANENT LOW-IMPEDANCE GROUND PATH. GROUNDING HARDWARE IN CLEAN ROOMS SHALL BE 316SS.
9. WHERE GROUNDING CONNECTIONS ARE MADE ON A PAINTED SURFACE, THE PAINT SHALL BE REMOVED TO EXPOSE A CLEAN METAL SURFACE AND COATED WITH ZINC-RICH, COLD GALVANIZING PAINT BEFORE THE CONNECTION IS MADE. GROUNDING CONNECTIONS SHALL NOT BE PAINTED OTHERWISE.
10. GROUNDING ROD TEST WELLS SHALL NOT BE LOCATED UNDER ROADWAYS, STORAGE AREAS OR ANY OTHER AREA WHERE THEY MAY BE SUBJECT TO PHYSICAL DAMAGE OR MAY BE RENDERED INACCESSIBLE.
11. ISOLATED, SENSITIVE ELECTRONIC GROUNDING ELECTRODE SYSTEMS SHALL BE BONDED TO OTHER GROUNDING ELECTRODES ON THE PREMISES AT ONLY ONE LOCATION AS INDICATED ON THE DRAWINGS.
12. FOR INDIVIDUAL CONDUCTORS IN CONDUIT, A SEPARATE, INSULATED EQUIPMENT GROUNDING CONDUCTOR SHALL BE INSTALLED IN EACH RACEWAY. METAL CONDUIT AND EMT ALONE SHALL NOT BE RELIED UPON AS THE SOLE EQUIPMENT GROUNDING CONDUCTOR.
13. MULTICONDUCTOR CABLE ASSEMBLIES SHALL CONTAIN AN EQUIPMENT GROUNDING CONDUCTOR OR ONE OF THE CONDUCTORS SHALL BE IDENTIFIED AS SUCH WITH GREEN MARKING TAPE AND HEAT SHRINK TUBING MARKED AS "GND" (GREEN).
14. WHERE MULTICONDUCTOR CABLE IS INSTALLED IN A CABLE TRAY AND THE EQUIPMENT GROUNDING CONDUCTOR WITHIN THE CABLE ASSEMBLY DOES NOT COMPLY WITH THE NEC MINIMUM REQUIRED SIZE, A SEPARATE INSULATED, PROPERLY SIZED, COPPER EQUIPMENT GROUNDING CONDUCTOR SHALL BE INSTALLED WITHIN THE CABLE TRAY AND CONNECTED AT EACH END TO THE EQUIPMENT GROUNDING CIRCUIT CONDUCTORS IN THE CABLE.

LIGHTING

1. FOR LIGHTING FIXTURE SCHEDULE SEE DRAWINGS LP100-EJ-00000004 AND - 00000005.
2. FOR GENERAL STANDARD LIGHTING INSTALLATION DETAILS, SEE DRAWINGS LP100-EJ-00000201 THRU -00000206. CUSTOM, NODE-SPECIFIC DETAILS SUPPLEMENT THE STANDARD LIGHTING DETAILS.
3. FOR ILLUMINATION (FOOT CANDLE) LEVELS AND ADDITIONAL LIGHTING INFORMATION SEE DESIGN CRITERIA DOCUMENT LP100-265-DBD-001. SEE SPECIFICATION LP100-265-EMS-16500 FOR LED INTERIOR LIGHTING AND SPECIFICATION LP100-265-EMS-16501 FOR LED EXTERIOR LIGHTING.
4. LIGHTING FIXTURES (LUMINAIRES) SHALL BE INSTALLED PER THE MANUFACTURER'S INSTRUCTIONS AND SHALL BE SUITABLE FOR THE PURPOSE.
5. LIGHTING CIRCUITS RATED MORE THAN 120 VAC SHALL BE SUPPLIED FROM DEDICATED 480Y/277V LIGHTING DISTRIBUTION PANELBOARDS. BRANCH CIRCUIT BREAKER RATINGS ARE INDICATED ON THE PANEL SCHEDULES.
6. LUMINAIRES AND LIGHTING CIRCUITS SHALL BE ARRANGED SUCH THAT A FAILURE IN ONE BRANCH CIRCUIT DOES NOT LEAVE AN AREA IN TOTAL DARKNESS WITH A MIN. OF 1 FOOTCANDLE FOR EGRESS SUPPLIED BY THE EMERGENCY LIGHTING INVERTER.
7. COMMON NEUTRALS SHALL NOT BE USED. EACH 120 OR 277 VAC LUMINAIRE BRANCH CIRCUIT SHALL HAVE AN INDIVIDUAL NEUTRAL CONDUCTOR.
8. BOTTOM OF FIXTURE (BOF) FIXTURE ELEVATIONS INDICATED FOR LIGHTING FIXTURES (LUMINAIRES) ARE REFERENCED FROM TOP OF GRADE, FINISHED FLOOR, OR PLATFORM, UNLESS NOTED OTHERWISE.
9. LUMINAIRES SHALL BE LOCATED TO AVOID CONFLICT WITH DUCTWORK, PIPING, STEEL STRUCTURES, CABLE TRAY, EQUIPMENT , OR OTHER ITEMS WHICH COULD CAUSE SHADOW AREAS.
10. SPLICES SHALL NOT BE MADE IN CONDUIT BODIES SUCH AS LB'S, T'S, LL'S, ETC., EXCEPT WHERE APPROVED OR INDICATED ON THE DRAWINGS, ALL BODIES/FITTINGS CONTAINING SPLICES SHALL BE SIZED (MIN.) PER NEC ART. 314.
11. FLOODLIGHTS SHALL BE AIMED IN THE DIRECTION INDICATED ON THE DRAWINGS. FINAL POSITIONING SHALL BE DETERMINED BY LILLY AFTER CONSTRUCTION IS COMPLETE AND SHALL BE PERFORMED BY THE ELECTRICAL CONTRACTOR.
12. LIGHTING BRANCH CIRCUIT CONDUCTORS SHALL BE IDENTIFIED AT EACH TERMINATION AND JUNCTION BOX WITH PRINTED HEAT SHRINK SLEEVES PER DWG. LP100-EJ-00000010.
13. EMERGENCY LIGHTING PANELS SHALL BE SUPPLIED BY AN EMERGENCY LIGHTING INVERTER LOCATED IN EACH BUILDING. EMERGENCY LIGHTING CIRCUITS SHALL BE INSTALLED IN SEPARATE CABLE TRAY SYSTEMS AND RACEWAYS THAN NON-EMERGENCY LIGHTING OR POWER CIRCUITS.
14. SPARE CONDUCTORS SHALL BE TERMINATED AND MARKED SPARE AS INDICATED ON THE DRAWINGS. IF NOT TERMINATED, THEN SPARE CONDUCTORS SHALL BE NEATLY COILED AND TAPED AT THE BOTTOM OF THE ENCLOSURE AND CLEARLY MARKED AS SPARE (SP).

CABLE TRAY

1. FOR STANDARD CABLE TRAY INSTALLATION DETAILS, SEE LP100-EJ-00000400 DRAWING SERIES.
2. THE CABLE TRAY SYSTEM SUPPORT LOCATIONS ARE BASED ON B-LINE NEMA 20C+ HEAVY DUTY (HD), ALUMINUM LADDER TRAY. DEVIATIONS ARE NOT PERMITTED UNLESS PRIOR APPROVAL IS OBTAINED FROM ENGINEERING AND CONTRACTOR ASSUMES RESPONSIBILITY FOR ADDITIONAL TRAY SUPPORTS REQUIRED.
3. HD METAL CABLE TRAYS SHALL BE SUPPORTED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. IF NON-HD AL TRAYS ARE USED, NEMA BI 50015 SUPPORT REQUIREMENTS SHALL APPLY WITHOUT EXCEPTION.
4. CABLE TRAY SPLICES AND MID-SPAN EXPANSION PLATES SHALL BE LOCATED AT QUARTER-SPAN, UNLESS NOTED OTHERWISE OR PERMITTED BY THE CABLE TRAY MANUFACTURER.
5. CABLE TRAY HARDWARE (NUTS, BOLTS, WASHERS, ETC.) SHALL BE 316SS UNLESS NOTED OTHERWISE. ALTERNATIVE MATERIALS DESCRIBED IN GENERAL NOTE 13, THIS DWG., ARE NOT APPLICABLE.
6. INDOOR BOTTOM OF CABLE TRAY (BOT) ELEVATIONS INDICATED ARE MEASURED FROM FINISHED FLOOR (INDOORS). OUTDOOR (BOT) ELEVATIONS INDICATE DISTANCE FROM TOP OF PIER LEVEL "EL -3'-0" T/PR PIER" UNLESS NOTED OTHERWISE.
7. CABLE DROP OUT FITTINGS SHALL BE USED WHERE CABLE EXITS THE TRAY THROUGH THE RUNGS. TRAY RUNGS MAY BE REMOVED WHERE REQUIRED FOR MINIMUM CABLE RADIUS BENDS AT DROP OUT FITTINGS.
8. FOR HORIZONTAL TRAY RUNS, CABLES SHALL BE SECURED TO THE RUNGS AT MAXIMUM INTERVALS OF 3'-0". FOR VERTICAL AND SLOPED TRAY RUNS, CABLES SHALL BE TIED AT MAXIMUM INTERVALS OF 2'-0". UV RESISTANT CABLE TIES SHALL BE USED IN ALL CASES.
9. SPLICING OF CABLES IN CABLE TRAY IS PROHIBITED WITHOUT OBTAINING PRIOR ENGINEERING APPROVAL.
10. EXPANSION JOINTS SHALL BE INSTALLED IN STRAIGHT, CONTINUOUS TRAY RUNS AS INDICATED ON THE DWGS. OR AT INTERVALS NOT LESS THAN RECOMMENDED BY THE MANUFACTURER'S INSTRUCTIONS. THE QTY. AND SPACING BETWEEN THEM ARE BASED ON THE MINIMUM AND MAXIMUM METAL CABLE TRAY SURFACE TEMPERATURES, NOT THE MIN./MAX. AMBIENT TEMPERATURES. SEE NOTE 12 AND 13 BELOW. HOWEVER, TRAY GAP SETTINGS ARE BASED ON THE AMBIENT TEMPERATURE AT THE TIME OF INSTALLATION AFTER CHARTING THE MIN./MAX. METAL TEMPERATURES PER DETAIL EL150A, DWG. LP100-EL-00000404.
11. AN ALUMINUM METAL TEMPERATURE OF 150 DEG. F (MAX.) SHALL BE USED FOR OUTDOOR ALUMINUM TRAYS EXPOSED TO DIRECT SUNLIGHT. ALUMINUM METAL TEMPERATURE OF 130 DEG. F (MAX.) SHALL BE USED FOR OUTDOOR ALUMINUM CABLE TRAYS NOT EXPOSED TO DIRECT SUNLIGHT.
12. A MINIMUM ALUMINUM METAL TEMPERATURE OF -20 DEG. F SHALL BE USED FOR OUTDOOR ALUMINUM METAL CABLE TRAYS.
13. FOR ALUMINUM METAL CABLE TRAY INDOOR INSTALLATIONS WHERE PERMANENT ENVIRONMENTAL CONTROLS ARE NOT IN OPERATION, APPLY NOTES 11 AND 12 ACCORDINGLY.
14. MOVED NOTE TO CONDUCTORS, GENERAL, THIS DWG.
15. ELECTRICAL CONTINUITY SHALL BE MAINTAINED BETWEEN DISCONTINUOUS SECTIONS OF METALLIC CABLE TRAY WITH THE USE OF APPROPRIATELY SIZED COPPER BONDING JUMPERS.
16. BONDING JUMPERS AT EXPANSION JOINTS AND BETWEEN DISCONTINUOUS SEGMENTS OF METALLIC CABLE TRAY SHALL BE RATED FOR THE MAXIMUM CURRENT RATING OF THE TRAY, UNLESS NOTED OTHERWISE. ALUMINUM 20C CABLE TRAY HAS A MAXIMUM AMPACITY OF 2000 A.
17. METALLIC CABLE TRAYS SHALL BE BONDED TO THE STRUCTURAL STEEL AT INTERVALS NOT TO EXCEED 40 FEET AS INDICATED ON DRAWINGS.
18. CABLE TRAYS SHALL BE IDENTIFIED AND MARKED PER DWG. LP100-EJ-00000414.
19. VENTILATED COVERS SHALL BE INSTALLED ON HORIZONTAL CABLE TRAYS LOCATED IN AREAS WHERE FALLING DEBRIS OR OBJECT MAY DAMAGE CABLES SUCH AS UNDERNEATH GRATED PLATFORMS, AND ON VERTICAL TRAYS UP TO A HEIGHT OF 8 FT. (MIN.) MEASURED FROM FINISHED SURFACE, (GRADE, PAVING, TOP OF PLATFORMS) OR ACCESSIBLE TO PEDESTRIAN OR VEHICULAR TRAFFIC.

CABLE CHANNEL

1. FOR STANDARD CABLE CHANNEL INSTALLATION DETAILS, SEE LP100-EJ-00000400 DRAWING SERIES.
2. CABLE CHANNELS SHALL BE INSTALLED PER MANUFACTURER'S INSTRUCTIONS AND LILLY SPECIFICATIONS.
3. CABLE CHANNEL MAY BE INDICATED DIAGRAMMATICALLY ON THE DRAWINGS. CABLE CHANNEL AND ASSOCIATED SUPPORTS SHALL BE DESIGNED AND ROUTED BY THE ELECTRICAL CONTRACTOR AND COORDINATED WITH OTHER TRADES TO AVOID INTERFERENCES.
4. CABLE CHANNEL SHALL BE VENTILATED, ALUMINUM, 1-3/4" D (MIN.), UNLESS NOTED OTHERWISE.
5. CABLE CHANNEL SHALL BE INSTALLED WITH 316 STAINLESS STEEL HARDWARE. ALTERNATIVE MATERIALS DESCRIBED IN GENERAL NOTE 13, THIS DWG., ARE NOT APPLICABLE.
6. CABLE CHANNEL SHALL BE SUPPORTED PER THE MANUFACTURER'S INSTRUCTIONS AT A MAXIMUM DISTANCE OF EVERY 10'-0" HORIZONTALLY AND 5'-0" VERTICALLY, AND WITHIN 6 IN. OF EACH END OF THE SEGMENT.
7. EACH HORIZONTAL CABLE SHALL BE SECURED TO THE CHANNEL EVERY 3'-0" (MAX.) FOR VERTICAL AND SLOPED CHANNEL RUNS, CABLES SHALL BE SECURED EVERY 2'-0" (MAX.) UV RESISTANT CABLE TIES SHALL BE USED IN ALL CASES.
8. CABLE CHANNEL SHALL BE BONDED PER THE NATIONAL ELECTRICAL CODE AND MANUFACTURER'S INSTRUCTIONS. ELECTRICAL CONTINUITY SHALL BE MAINTAINED AT DISCONTINUOUS SEGMENTS OF CABLE CHANNEL, CABLE TRAY AND CABLE CHANNEL, OR CABLE CHANNEL AND CONDUIT.
9. CABLE CHANNEL SHALL BE UTILIZED FOR ALL 600 V CABLES WITH A MAXIMUM CABLE SIZE OF 7/C, 10 AWG, AND FILLED NOT TO EXCEED THE WIDTH OR HEIGHT OF THE CHANNEL.
10. VENTILATED COVERS SHALL BE INSTALLED ON CABLE CHANNELS LOCATED IN AREAS WHERE SUBJECT TO DAMAGE BY FALLING DEBRIS OR IN LOCATIONS WHERE READILY ACCESSIBLE TO PERSONNEL.

UNDERGROUND CONDUIT

1. LONG RADIUS RIGID GALVANIZED STEEL ELBOWS SHALL BE USED UNDERGROUND. WHERE CONDUITS STUB-UP ABOVEGROUND, THEY SHALL EXIT IN THE VERTICAL PLANE WITH NO PART OF THE ELBOW RADIUS EXPOSED.
2. UNLESS NOTED OTHERWISE, UNDERGROUND CONDUIT ELBOWS SHALL BE EITHER:
   A. RIGID GALVANIZED STEEL FOR POWER OR CONTROL CONDUCTORS, OR
   B. REINFORCED THERMOSETTING RESIN (TYPE RTRC) FOR FIBER AND COMMUNICATION CIRCUITS.
3. LOW VOLTAGE CONDUIT DUCT BANKS SHALL BE INSTALLED 24" (MIN.) BELOW GRADE MEASURED FROM TOP OF CONCRETE TO THE TOP OF THE UNDERGROUND CONDUIT, AND ALSO COMPLY WITH NOTE 1 ABOVE, UNLESS NOTED OTHERWISE.
4. MV CONDUIT DUCT BANKS SHALL BE INSTALLED A MINIMUM OF 36" BELOW GRADE MEASURED FROM THE TOP OF CONCRETE AND ALSO COMPLY WITH NOTE 1 ABOVE, UNLESS NOTED OTHERWISE.
5. AT ROAD CROSSINGS, THE TOP OF THE REINFORCED CONCRETE ENCASEMENT SHALL BE A MINIMUM OF 42" BELOW THE ROAD SURFACE, UNLESS NOTED OTHERWISE.
6. ALL UNDERGROUND CONDUITS SHALL BE INSTALLED WITH A MINIMUM AMOUNT OF BENDS NOT TO EXCEED 270 DEGREES TOTAL, UNLESS NOTED OTHERWISE OR PRIOR ENGINEERING APPROVAL IS OBTAINED.
7. UNDERGROUND CONDUIT SHALL NOT BE INSTALLED IN THE 45° DEG. CONE OF DEFLECTION FROM THE LOWEST POINT OF CONCRETE FOUNDATIONS.
8. UNDERGROUND CONDUITS SHALL BE IDENTIFIED WITH TAGS PER LILLY SPECIFICATION LP100-265-EMS-16195 WHERE THEY EMERGE ABOVEGROUND. ALSO SEE DWG. LP100-EJ-00000010.
9. SPARE UNDERGROUND CONDUITS SHALL BE CAPPED IN AN APPROVED MANNER WITH SUITABLE, MANUFACTURED PERMANENT CAPS TO PREVENT ENTRANCE OF DIRT, WATER, OR OTHER DEBRIS. SPARE CONDUIT CAPS SHALL BE PAINTED AND MARKED PER LILLY SPECIFICATION LP100-265-EMS-16112.
10. RIGID METAL CONDUIT STUBBED UP IN CONCRETE SHALL HAVE 2-10 MIL LAYERS OF 3M SCOTCH-RAP 50 CORROSION PROTECTION APPLIED WHERE THE CONDUIT EXITS THE CONCRETE PRIOR TO THE POUR TO REDUCE CORROSION.
11. FOR MORE UNDERGROUND REQUIREMENTS, SEE GENERAL CONDUCTOR NOTES, THIS DWG.

BATTERIES

1. BATTERY INSTALLATIONS SHALL COMPLY WITH THE NEC, LOCAL CODES AND STANDARDS, THE MANUFACTURER'S INSTRUCTIONS AND LILLY SPECIFICATIONS.
2. BATTERIES SHALL NOT BE STORED ON SITE PRIOR TO INSTALLATION AND SHALL BE INSTALLED DURING THE COMMISSIONING PHASE OF CONSTRUCTION.

MOTOR CONTROL STATIONS

1. MOTOR CONTROL STATIONS SUCH AS JOG-OFF-AUTO, STOP-START, ETC., LOCATED IN HAZARDOUS CLASSIFIED LOCATIONS, SHALL HAVE FACTORY SEALED CONTACTS, UNLESS NOTED OTHERWISE, AND SHALL BE IDENTIFIED FOR THE LOCATION.
2. MOTOR CONTROL STATIONS SHALL HAVE NAMEPLATES INSTALLED IN ACCORDANCE WITH LILLY SPECIFICATION LP100-265-EMS-16195.

RECEPTACLES & SWITCHES

1. FOR STANDARD SWITCH AND RECEPTACLE INSTALLATION DETAILS, SEE DRAWING LP100-EJ-00000601 THRU -00000608.
2. CONVENIENCE RECEPTACLES SHALL HAVE GROUND FAULT CIRCUIT INTERRUPTER PROTECTION PROVIDED IN ACCORDANCE WITH THE DRAWINGS AND THE NEC.
3. NOTE DELETED.

CONDUCTORS, GENERAL

1. THE ELECTRICAL CONTRACTOR SHALL PERFORM CABLE PULLING CALCULATIONS FOR ALL UNDERGROUND CONDUITS REGARDLESS OF LENGTH OR DEGREES OF BENDS AND FOR CABLE TRAY OR CONTINUOUS ABOVEGROUND CONDUIT RUNS EXCEEDING 200 FT. OR 270 DEGREES OF BENDS, UNLESS PRIOR WAIVER FROM ENGINEERING IS OBTAINED.
2. PROPER CABLE PULLING EQUIPMENT SHALL BE USED FOR CABLE INSTALLATION. DYNAMOMETERS, PULLEYS, ROLLERS, SHEAVES, TUGGERS, ETC. SHALL BE USED FOR PULLS OTHER THAN THOSE DONE MANUALLY. MAXIMUM CABLE TENSIONS DURING PULLS SHALL NOT EXCEED THE CABLE MANUFACTURERS RECOMMENDATIONS. MANUFACTURED PULLING LUBRICANT SUCH AS POLYWATER J SHALL BE USED FOR CABLE PULLS IN CONDUIT.
3. AN AMBIENT TEMPERATURE OF 40 °C (104 °F) (MAX.) SHALL BE ASSUMED FOR CONDUCTOR AMPACITY CALCULATIONS FOR ALL CONDUCTORS LOCATED ABOVEGROUND.
4. UNDERGROUND CONDUCTOR AMPACITIES, INCLUDING THE AMBIENT TEMPERATURE USED SHALL BE CALCULATED BASED ON THE ARRANGEMENT OF THE DUCT BANK AND SITE SPECIFIC SOIL DATA WITH APPROPRIATE SOFTWARE.
5. CABLE/WIRE CONNECTORS, LUGS, TERMINALS, INCLUDING IN-LINE SPLICES, SHALL BE NRTL LISTED, CU, 90 DEG. C, COMPRESSION-TYPE, WITH HOT OR COLD SHRINK INSULATION RATING EQUAL TO OR EXCEEDING THE CONDUCTOR INSULATION. TERMINALS PART OF NRTL LISTED OR FIELD CERTIFIED MANUFACTURED EQUIPMENT (I.E., PART OF BREAKERS, MCCS, SWITCHGEAR, DISTRIBUTION BLOCKS, ETC.), ARE EXCLUDED FROM THIS REQUIREMENT.
6. CABLES ROUTED IN CABLE TRAYS SHALL BE IDENTIFIED AT EACH END OF THE CABLE RUN AND SHALL COMPLY WITH DWGS. LP100-EJ-00000002 AND -00000010, UNLESS NOTED OTHERWISE.
   A. FOR FEEDERS 3 AWG AND LARGER USE 3/4" BY 1.5 INCH NYLON MARKER PLATES AFFIXED WITH NYLON TIE WRAPS AT BOTH ENDS.
   B. FOR CONTROL WIRING, BRANCH CIRCUITS AND FEEDER WIRING LESS THAN 3 AWG USE COMPUTER PRINTABLE SLIP ON, HEAT SHRINKABLE WIRE MARKERS AT BOTH ENDS.

CONDUCTORS AND TERMINATIONS > 2000 V

1. MEDIUM VOLTAGE (MV) IS CONSIDERED TO BE 2000 V, OR GREATER, FOR THE LP1 PROJECT.
2. NOTE DELETED AND MOVED TO CONDUCTORS, GENERAL.
3. MV MULTICONDUCTOR CABLES INSTALLED IN CABLE TRAY SHALL BE TYPE MC OR MC-HL AS INDICATED ON THE CABLE SCHEDULES AND SHALL BE CABLE TRAY RATED. MV TYPES MC OR MC-HL CABLE SHALL NOT BE INSTALLED IN CONDUITS OR OTHER RACEWAYS.
4. INDIVIDUAL MV CONDUCTORS ARE PERMITTED TO BE INSTALLED IN UNDERGROUND OR ABOVEGROUND CONDUIT OR CABLE TRAY, UNLESS NOTED OTHERWISE. WHERE INSTALLED OUTDOORS IN CABLE TRAY, THE CONDUCTORS SHALL BE RATED FOR CABLE TRAY USE AND HAVE UV RESISTANT INSULATION OR OVERALL COVERING.
5. MV FEEDER AND BRANCH CIRCUIT CONDUCTORS, (OTHER THAN MOTOR BRANCH CIRCUITS SUPPLIED FROM AN ADJUSTABLE SPEED DRIVE (ASD) OR VARIABLE FREQUENCY DRIVE (VFD), SHALL BE TYPE EPR, 133% RATED FOR THE VOLTAGE, 105 °C INSULATION (MIN.), AND A 10 MIL COPPER TAPE SHIELD. UNLESS NOTED OTHERWISE.
6. MV MOTOR BRANCH CIRCUIT CONDUCTORS ON 5 KV SYSTEMS DERIVED FROM AN ASD OR VFD SHALL BE TYPE EPR, 100% RATED FOR 15 KV AND 105 °C INSULATION (MIN.), AND A 10 MIL COPPER TAPE SHIELD, UNLESS NOTED OTHERWISE. IN LIEU OF THIS REQUIREMENT, 173% RATED 5/8 KV MULTICONDUCTORS SHALL BE PERMITTED.
7. MV CABLES SHALL BE MARKED IN ACCORDANCE WITH DWG. LP100-EJ-00000010 AND LILLY SPECIFICATION LP100-265-EMS-16195.

CONDUCTOR COLOR CODING FOR ALTERNATING CURRENT SYSTEMS , 60 Hz, > 2000 V

CONDUCTOR	COLOR
PHASE A	BLACK
PHASE B	RED
PHASE C	BLUE
GROUNDING/ EARTHING	GREEN OR BARE (IF BARE MARKING NOT REQ'D)

CONDUCTORS AND TERMINATIONS, 600 V (MAX.)

1. 600 VOLT CONDUCTORS SHALL HAVE TYPE XHHW-2 OR XLPE INSULATION RATED 90 DEG. C FOR WET LOCATIONS, UNLESS NOTED OTHERWISE. MULTICONDUCTOR CABLES SHALL BE TYPES TC-ER OR MC AS INDICATED ON THE CONDUIT/CABLE SCHEDULES.
2. FOR MOTORS SUPPLIED BY AN ADJUSTABLE SPEED DRIVE (ASD), SHIELDED VFD RATED MULTICONDUCTOR CABLES TYPES TC-ER OR MC, XHHW-2 INSULATED CABLE SHALL BE INSTALLED FROM THE ASD TO THE MOTOR, UNLESS NOTED OTHERWISE.
3. ALL WIRE AND CABLE SHALL HAVE WIRE MARKERS AND TAGS PER DWG. LP100-EJ-00000010 AND LILLY SPECIFICATION LP100-265-EMS-16195.
4. MULTICONDUCTOR CABLES CONTAINING BLACK, NUMBERED CONDUCTORS (ALL SIZES), OR BLACK, INDIVIDUAL CONDUCTORS WITHIN A CONDUIT FOR POWER FEEDER CIRCUITS, 4 AWG AND LARGER, SHALL MARKED AND IDENTIFIED WITH THE FOLLOWING COLOR SCHEME:

COLOR CODING FOR 600 V, 60 HZ CONDUCTORS

CONDUCTOR	208Y / 120V	480Y / 277V
PHASE A	BLACK	BROWN
PHASE B	RED	ORANGE
PHASE C	BLUE	YELLOW
NEUTRAL	WHITE	GRAY
GROUNDING/ EARTHING	GREEN	GREEN
ISO GROUNDING/ EARTHING	GREEN W/ YELLOW STRIPE	GREEN W/ YELLOW STRIPE

5. INDIVIDUAL GROUNDED CONDUCTORS 6 AWG AND SMALLER IN RACEWAYS SHALL BE WHITE OR GRAY THE ENTIRE LENGTH OF THE CONDUCTOR. INDIVIDUAL INSULATED GROUNDING CONDUCTORS IN RACEWAYS SHALL BE GREEN OR GREEN WITH A YELLOW STRIPE(S) THE ENTIRE LENGTH OF THE CONDUCTOR.
6. MINIMUM CONDUCTOR SIZES SHALL BE AS FOLLOWS:

   14 AWG FOR MOTOR CONTROL AND CONTROL PANEL WIRING
   12 AWG FOR GENERAL POWER WIRING
   12 AWG FOR LIGHTING SYSTEM WIRING
   12 AWG FOR LIGHTING SYSTEM HOME RUNS
   10 AWG FOR CURRENT TRANSFORMER CIRCUITS

7. AMPACITIES OF CONDUCTORS RATED 600 V, NOMINAL, OR LESS, SHALL BE PERMITTED TO BE BASED ON THE 90° C INSULATION RATING INDICATED IN NEC TABLE 310.16 FOR XHHW-2 INSULATION. HOWEVER, THE FINAL AMPACITY FOR THE OVERCURRENT PROTECTION DEVICE SHALL NOT EXCEED THE 75 °C AMPACITIES OF NEC TABLE 310.16 AFTER AMBIENT TEMPERATURE ADJUSTMENTS AND OTHER DERATING FACTORS HAVE BEEN APPLIED.
8. SPRING TYPE (WIRE NUT) CONNECTORS ARE PROHIBITED, UNLESS NOTED OTHERWISE.
9. PIN TERMINAL OR RING-TONGUE CONNECTORS SHALL BE USED TO TERMINATE SMALL CONDUCTORS 14 AWG THROUGH 10 AWG.
10. IN-LINE SPLICES SHALL BE NOT BE PERMITTED UNLESS PRIOR ENGINEERING APPROVAL IS OBTAINED. WHERE IN-LINE SPLICES ARE APPROVED, INSULATED IRREVERSIBLE COMPRESSION TYPE BUTT SPLICES SHALL BE USED AND CONTRACTOR SHALL MARK LOCATIONS ON A SET OF RED-LINED AS-BUILT DRAWINGS. RED-LINED DRAWINGS SHALL BE SUBMITTED TO OWNER.
11. MECHANICAL SPLICES (POLARIS TAPS) ARE PROHIBITED.
12. MOTOR TERMINATIONS, 600 V, NOMINAL, OR LESS, SHALL BE COMPRESSION LUGS WITH BOLTS, WASHERS AND LOCKWASHERS INSULATED WITH RAYCHEM GELCAPS, OR APPROVED EQUAL. SPLIT-BOLT AND WIRE NUT TYPE CONNECTORS ARE PROHIBITED UNLESS OTHERWISE NOTED.

RAYCHEM GELCAP SELECTION CHART

PART NO.	CONDUCTOR SIZE
GELCAP 1	14 - 10 AWG
GELCAP 2	8 - 2 AWG
GELCAP 3	2 - 4/0 AWG
GELCAP 4	250 - 500 KCMIL

13. WHERE CONDUCTORS HAVE BEEN INCREASED IN SIZE FOR VOLTAGE DROP, TEMPERATURE ADJUSTMENTS, OR OTHER DERATING, NRTL LISTED, CU, COMPRESSION PIN ADAPTERS ARE PERMITTED IN LIEU OF TRANSITION BOXES AT THE SOURCE OR LOAD END OF THE CIRCUIT PROVIDED ALL OF THE FOLLOWING CONDITIONS ARE MET:
    A. THE CONDUCTOR INSULATION DOES NOT EXCEED 600 V, AND
    B. THE WIRE FILL WITHIN THE ENCLOSURE IS ADEQUATE, AND
    C. THE CONDUCTOR DEFLECTION AT THE TERMINAL IS IN ACCORDANCE WITH NEC ART. 312, AND
    D. THE CABLE IS NOT DAMAGED AND DOES NOT EXCEED THE MINIMUM BENDING RADIUS PERMITTED, AND
    E. THE COMPLETED INSTALLATION IS IN ACCORDANCE WITH THE NEC, AND
    F. THE CONTRACTOR FIELD MARKS THE CONDUIT/CABLE SCHEDULE AS-BUILT SET OF DRAWINGS WHERE ADAPTER PINS ARE USED, AND
    G. THE PIN ADAPTERS ARE INSULATED WITH THE MANUFACTURER'S PROVIDED INSULATION KIT.

SPECIFICATION REFERENCES

NUMBER	TITLE
LP100-265-EMS-15170	ELECTRICAL HEAT TRACING
LP100-265-EMS-16010	GENERAL PROVISIONS
LP100-265-EMS-16080	ELECTRICAL ACCEPTANCE TESTING
LP100-265-EMS-16100	BASIC ELECT MATERIAL AND METHODS
LP100-265-EMS-16110	CABLE TRAYS
LP100-265-EMS-16112	UNDERGROUND ELECTRICAL INSTALLATION
LP100-265-EMS-16115	LIGHTNING AND STATIC PROTECTION
LP100-265-EMS-16123	3/C MEDIUM VOLTAGE POWER CABLES
LP100-265-EMS-16124	1/C MEDIUM VOLTAGE POWER CABLES
LP100-265-EMS-16125	LOW VOLTAGE POWER CABLES
LP100-265-EMS-16154	ELECTRIC MOTORS THROUGH 400 HP
LP100-265-EMS-16155	AC ADJUSTABLE SPEED DRIVES
LP100-265-EMS-16165	MOTOR CONTROL
LP100-265-EMS-16169	MOTOR CONTROL CENTERS
LP100-265-EMS-16195	ELECTRICAL IDENTIFICATION
LP100-265-EMS-16300	NATURAL GAS-ENGINE-DRIVEN GENERATOR SET
LP100-265-EMS-16319	LV UNIT SUBSTATIONS
LP100-265-EMS-16320	PDC (E-HOUSE)
LP100-265-EMS-16330	AUTOMATIC TRANSFER SWITCH (ATS)
LP100-265-EMS-16436	MEDIUM-VOLTAGE METAL-CLAD SWITCHGEAR (5-15kV)
LP100-265-EMS-16460	DRY TYPE TRANSFORMERS THROUGH 225 KVA
LP100-265-EMS-16462	NON-SEGREGATED PHASE BUS DUCT
LP100-265-EMS-16464	LIQUID IMMERSED POWER TRANSFORMER
LP100-265-EMS-16465	PAD MOUNTED TRANSFORMER
LP100-265-EMS-16470	PANELBOARDS
LP100-265-EMS-16500	LED INTERIOR LIGHTING
LP100-265-EMS-16501	LED EXTERIOR LIGHTING
LP100-265-EMS-16510	ADDRESSABLE LUMINAIRE LIGHTING CONTROLS
LP100-265-EMS-16520	LIGHTING CONTROL DEVICES
LP100-265-EMS-16530	LIGHTING INVERTER
LP100-265-EMS-16611	UPS SYSTEM (9 KW TO 450 KW)
LP100-265-EMS-16800	INTEGRATED POWER CENTER (IPC)
LP100-265-EMS-16995	ELECTRICAL REQUIREMENTS FOR PACKAGED EQUIPMENT

SAMUEL BENEFIELD
No. PE12300742
STATE OF INDIANA

Covering Rev 6, 4Feb2026

Samuel Benefield Digitally signed by Sam Benefield Date: 2026.02.04 10:00:34 -05'00'

REVISION 6 NOTES:

1. NOTE ADDED TO ABOVEGROUND CONDUIT CLARIFYING WHERE EMT IS PERMITTED.

6

REV	ZONE		PROJECT NUMBER / DESCRIPTION		CHANGE CONTROL	DATE
6	ALL		ISSUED FOR CONSTRUCTION		N/A	28-JAN-2026
5	ALL		ISSUED FOR CONSTRUCTION		N/A	20-NOV-2025
4	ALL		ISSUED FOR CONSTRUCTION		N/A	20-OCT-2025
3	ALL		ISSUED FOR CONSTRUCTION		N/A	08-OCT-2025
2	ALL		ISSUED FOR CONSTRUCTION		N/A	05-FEB-2025

REVISIONS

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EFDEE Engineering PC
PROJECT NO. B1LG
NODE: GEN

REV	DATE	DES	ENG	CHK	APP	PM APP
6	28-JAN-2026	EG	SB	MV	SB	
5	20-NOV-2025	EG	SB	MV	SB	
4	20-OCT-2025	EG	SB	MV	SB	
3	08-OCT-2025	EG	SB	MV	KSW	
2	05-FEB-2025	EG	SB	MV	KSW	

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ELECTRICAL
GENERAL NOTES

SYSTEM TYPE: NA
EDIFICE/BUILDING: 00
LEVEL/FLOOR: 0
SUB DISCIPLINE: EJ
LOCATION CODE: LP1
LOCATION/GRID:
CCI CODE:
PROJECT NAME: LP1-306-1181
CRITICAL QUALITY
ENTERPRISE GROUP: API
DRAWING INTENTION:
DRAWING SCALE: NONE
CRITICAL BUSINESS
EQUIPMENT CLASSIFICATION: NA
ALTERNATE DOCUMENT ID: NA
MERIDIAN REV: 6
CRITICAL HSE
VENDOR: NA
DOCUMENT ID: LP100-EJ-00000001
NON-CRITICAL
ENGINEERING AGENCY DOC ID: NA

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